

Exploring Decoupled Generation for Enhancing VR Interaction and Control

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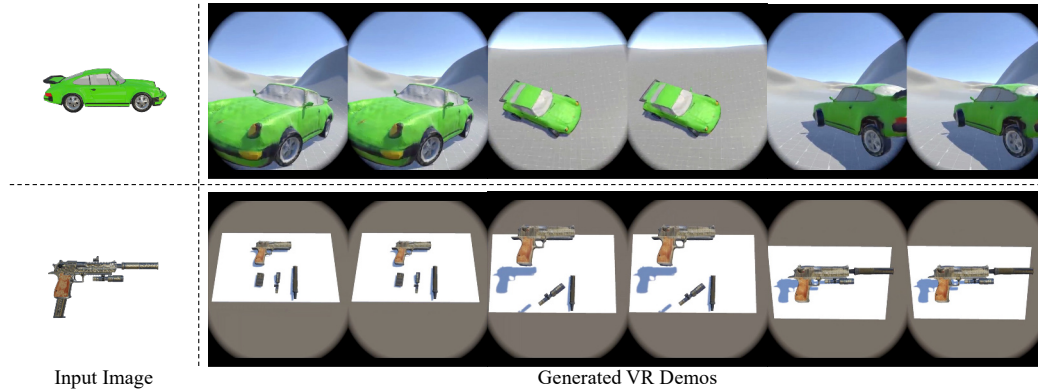


Figure 1: Virtual reality (VR) demo of our generated 3D objects with decoupled components. Firstly, we utilize a semantic segmentation model to give object components. Then we use a diffusion model to generate the multi-view information, which is further adapted for 3D reconstruction. Finally, the 3D components are assembled and transformed based on user preferences. The result showcases accurate reconstruction and flexible manipulation in VR, validating the effectiveness of our Decoupled3D framework.

ABSTRACT

This paper presents Decoupled3D, a novel framework for 3D object generation with decoupled components. Decoupled3D uniquely decouples and optimizes object component meshes from a single image, producing not only an overall shape for the object but also the independent modeling for object components. Decoupled3D serves as a plug-and-play module, which is applicable to various object images and existing 3D reconstruction methods, offering high flexibility and scalability. Experimental results suggest the potential of digital creation and virtual reality applications. The survey results indicate that participants believe our proposed Decoupled3D offers superior control and design capabilities.

Index Terms: Image to 3D, 3D Reconstruction, 3D Generation, Diffusion, VR, AR

1 INTRODUCTION

Despite the rapid advancement in 3D object generation and reconstruction technology in recent years [7, 5], the direct application of AI-produced objects in industries such as games, film, and virtual reality remains limited. This is mainly due to the absence of accurate modeling on decoupled object components.

Typically, existing single-view 3D reconstruction methods, such as SyncDreamer [3], One-2-3-45 [1], and Zero123++ [6], fail to identify object components, where objects are considered as rigid bodies without any component decomposition. This limitation becomes apparent when dealing with complex objects, in which separate manipulation of individual components is required to accom-

modate various scenarios, such as manipulating car wheels. As a result, the 3D objects produced by these methods still fall short when it comes to real-world applications.

To resolve such challenge, we investigate a novel task of generating objects with decoupled components. This task would play a crucial role in closing the gap between the AI generated 3D objects and the downstream applications such as game design and file creation, since it enables the direct creation of operable components and enhances the usability of generated objects. Specifically, we aim to 1) achieve 3D object generation from monocular image input and 2) maintain the independence and details of 3D object components, thereby offering greater flexibility and interactivity for the downstream tasks.

This paper presents Decoupled3D framework, discussing its performance across various application scenarios, and validates its effectiveness using real-world data. As shown in Fig. 1, we demonstrate the decoupled generation for 3D objects, e.g. cars, where components such as wheels are individually generated. Then, the 3D components can be spatially controlled independently when assembled in a game engine. Decoupled3D exhibits exceptional capability in generating high-quality, detail-rich 3D objects from limited viewpoints, while also enhancing texture fidelity, setting a new standard for 3D modeling.

2 METHODOLOGY

Our framework comprises three key stages: (1) Segmentation Guidance and Component Isolation, (2) Multi-View Image Generation, and (3) Component-Level 3D Reconstruction and Assembly.

Firstly, we utilize a Multimodal Large Language Model (MLLM) to generate segmentation guidance points based on the contextual information in the input image. These points are used to instruct a segmentation model to isolate individual object components accurately [2]. This is crucial for fine-grained component segmentation where explicit guidance points are necessary, and we

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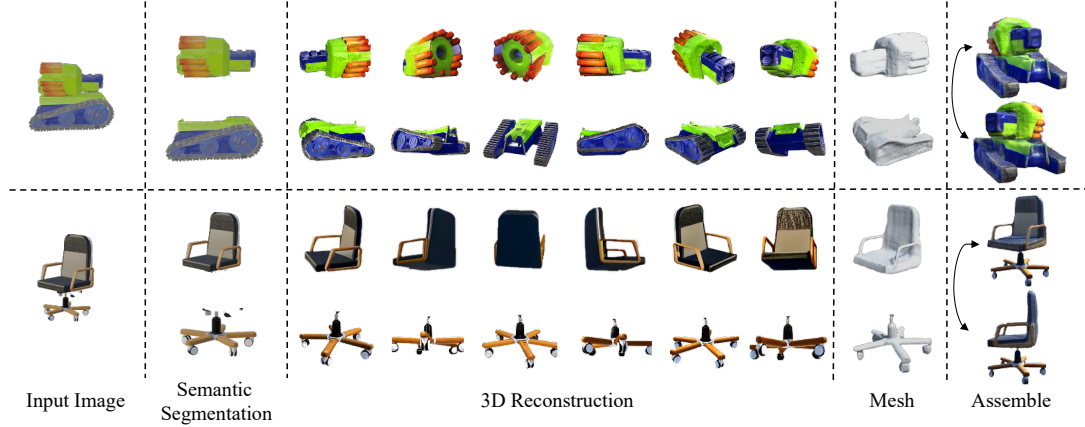


Figure 2: Illustration of the Decoupled3D framework results. The first image in the sequence is the input to the framework. Subsequent images show the segmentation results and the multi-view synthesis outputs. The final images display the extracted mesh and the assembled model.

leverage the visual understanding capabilities of MLLM to automate this process, avoiding a fully manual heuristic approach.

Secondly, for components with limited or occluded views, common challenges in single-image inputs, we employ state-of-the-art multi-view synthesis models One-2-3-4-5 [1], SyncDreameer [3], and InstantMesh[8] to generate consistent multi-view images, ensuring that each component can be reconstructed with sufficient detail. These models are modular and can be adapted to specific application requirements within our pipeline. The objective here is to utilize the learned priors of diffusion models to recover the geometric information of objects from a single view.

We integrate InstantNGP [4] for efficient training and inference, enabling high-quality neural surface reconstruction. Isosurface extraction techniques are employed to generate high-quality meshes from synthesized multi-view images, accelerating 3D geometry rendering while improving the accuracy and visual quality of the final 3D models through direct geometric supervision.

Finally, the reconstructed components are assembled in a virtual reality environment like Unity, facilitating complex animations and interactive behaviors.

3 RESULTS

As shown in Figure 2, we demonstrate the potential of the Decoupled3D framework to generate high-quality 3D models from a single-view input, illustrating its robustness and versatility. The results demonstrate excellent decoupled generation capabilities, with both the geometry and textures of the mesh proving sufficient for practical applications. Our results indicate that while the framework is adept at processing simpler shapes, the accurate reconstruction of complex geometries, such as intricate mechanical components or organic shapes, still poses a challenge. Future research should focus on developing a more end-to-end decoupled generation method, potentially integrating automated assembly and control assignment functionalities.

4 USER STUDY

We conducted a user study with 66 participants experienced to evaluate the advantages and limitations of part-based decoupled generation in VR design, comprising 56.06% males and 43.94% females. Professions included VR game designers (31.82%), VR application developers (30.3%), 3D modelers (18.1%), and students (19.7%). Experience in 3D design was categorized as less than 1 year (18.18%), 1-3 years (22.73%), 3-5 years (33.33%), and over 5 years (25.76%). Experiments were conducted remotely using participants' own setups compatible with Unity or Unreal Engine.

The user study involved two tasks to evaluate control precision and user perception in VR 3D applications. In Task 1, participants

evaluated holistic and part-based decoupled generation methods by implementing model control in a game engine, with a specific focus on part-level manipulation. In Task 2, they completed a survey to share preferences, ease of control, perceived benefits, and suggestions for improving the part-based decoupled generation approach.

Results show that 66.67% preferred decoupled generation for its flexibility, precision, and design freedom. However, 57.58% suggested improving generation speed, and 54.55% recommended enhancing automation. Despite these needs, 83.33% recognized its potential, and 46.97% were highly willing to adopt it, highlighting its promising future in VR, game design, and virtual environments.

4.1 Conclusion

This paper introduces the Decoupled3D framework, which is capable of decoupled part generation from single-view images with adjustment and editing capabilities. We have designed an architecture that integrates semantic segmentation models, multi-view diffusion models, and 3D reconstruction models. Experimental results and user studies show that the Decoupled3D framework offers greater potential, enhanced flexibility, interactivity, and fine control in animation, game design, and virtual reality.

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